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Icons Library Guidelines

This section will guide you through the process of creating and including your own graphic elements into the Reactome Icon Library and become a part of this growing community.

Since Reactome is a free and open-source project, the Reactome Library is under a Creative Commons license. As a result, if your creations are included in our library, they will follow the same policy. Feel free to use any tool or application that allows you to create **vector graphics** and, please, make sure that the following specifications are implemented in the files you will be creating.

Generic guidelines

- Keep simplicity in mind. The main goal for this Icon Library is to provide easy to read graphic components for our users and simplicity is key to this matter.
- Consistency is important for Reactome and the Icon Library. Please, before submitting a proposal, read the guidelines we have set for the different types of components we use.
- Each icon should represent a single element only. After receiving feedback we open this rule to receptors when binding other receptors in order to make it easy for our users.
- Every element should be placed into a separate file, **200** by **200px** in size, with **RGB colour mode** and **White background**.
- Use only **Arial Bold 12pt**. Text colour should be chosen depending on the text's position. In case the text is **inside** a coloured shape, its colour should be **White**, while in case the text is **outside** of a coloured shape, its colour should be **Black**.
- All text should be **Uppercased** except for words describing **greek** characters (e.g alpha, beta, gamma, etc.). In that case those words should be written in **Lowercase**. It should be noted that Greek or other Unicode characters must be avoided, as they are not supported for the moment.
- Your elements should only comprise vector graphics. Please make sure that you have not included any bitmap image in it. Including raster images often results in larger file sizes and thus negatively impacts software performance.
- You can place your new element in any of the seven categories of the Reactome Library; Cell elements, Cell types, Compounds, Human tissue, Ion channels, Proteins and Receptors.

Cell elements

Keeping simplicity is one of our targets so we recommend the use of **simple figures**, such as circles, squares and triangles as a base. For colouring we suggest the use of shades of **Primary Colours** (Red, Blue and Yellow) and **Secondary Colours** (Purple, Green and Orange).

Vector graphic applications allow you to use **Gradient tools** to blend two or more colours for the same shape. We recommend you to use these tools so you can create more appealing graphics. See the Mitochondrion below:



(a) Without the use of gradients, the Mitochondrion element looks plain while in (b) through the correct use of gradients the element looks more 3D-like, resulting in a more attractive and playful image.

Cell types

In order to ensure a sense of unity and a consistent look and feel, we suggest applying the same principles; simple shapes and colours.

For representing types of cells, please keep in mind that we are not using scale to portray the different organelles and elements inside the cell cytoplasm. Additionally, to maintain simplicity, try to avoid overloading the the overall design with too much detail. The following table features 3 examples:

	*This **Microbe** element is formed by a circle with a radial gradient of two colours. To represent the membrane, we have added an outline with a different gradient. To convey the idea of danger we use this squares as spikes and a red shade in their gradients.*
	*We have drawn this **Macrophage** element with shaky tentacles and we have used a not uniform outline (membrane) to give it a soft and flexible feeling.*
	*For this **Pathogen** element we used a large light gradient to represent a shiny and hard capsule. To show its death we draw holes on its surface and play with a darkened, sad colour.*

Compounds

Compounds are represented by very geometrical elements, following more specific guidelines.

Simple chemical elements or compounds are usually portrayed in the Library with their chemical element symbol (e.g. Ca for Calcium) or the name of the compound (e.g. IFN gamma) using **Arial bold 12pt** in **White** colour.

Simple chemical elements are represented by an **octagon** of **28px** height and two plain colours, one for the fill and a different one for the stroke (**2pt**). In case the text of the compound is too long to fit in the octagon, we suggest stretching the shape until it fits the name, respecting its lateral edges, like follows:



An example of a simple chemical element.

Complex compounds can be represented with a variety of shapes, always simple and easy to differentiate one from each other.

	We can use circles with or without outline.
	*We can use hexagons, in the same way as for the simple compounds, just adding a gradient on their stroke.*

Human tissue

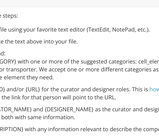
Human tissue elements are mainly used as backgrounds in our diagrams and the way of representing these organs depends on the illustrator's skills and taste. To ensure unity in the Reactome Library we suggest to keep these illustrations as simple as possible and visualise them as toys; so they convey a nice, soft feeling, with a plastic-like, very friendly and approachable look.



Examples of human tissue elements; Liver and Blood vessel.

Ion channels

These representations are quite simplified and, thus, elements are designed to look like small funnels that cross through the membrane, allowing elements to move from inside of the cell to outside and vice versa. There are hundreds of different ion channels and we suggest differentiating them with the use of colours in various combinations.



Examples of ion channels.

Proteins

There are tens of thousands of different proteins, some of them with specific shapes. Proteins in the Reactome Library follow the standard representation as rounded rectangle shapes. As a result, we suggest using the **Rounded Rectangle Tool** to draw a shape with **10px** radius for the corners and **20px** height for single line of text, or **30px** for a double line. The stroke should be **3pt** and we suggest choosing an irregular profile (in Adobe Illustrator, use the default **Width Profile 2**). For the name of the proteins we suggest to follow the same guides as before: **White Uppercase Arial Bold 12pt**.

The shape will contain the simplified name of the protein following the general instructions for Reactome Library. It should be filled with a simple gradient of two colours. One of the colours of this gradient should also be used as the stroke colour.



Examples of protein elements in their rounded rectangle representation.

In case a protein has a specific shape, we suggest keeping the standard representation as long as it is not too complicated. Otherwise, we suggest opting for a more simplified version.



Examples of protein elements represented in their standard representation.

Receptors

Like proteins, there is a huge number of receptors. In cases where there is a standard, but quite complex, way to represent them, we suggest adopting a more simplified version. If this is too difficult we suggest finding a more simple representation.

Examples of receptor elements

Metadata

You are encouraged to accompany your element with more information including a short description and extra details about its designer and curator. Upon inclusion of your element into the Reactome Library, all metadata information will be available to the public through our landing page.

In order to do so, you need to create an additional metadata file using your favorite text editor. The file should have the following structure:

```
<metadata>
    <categories>
        <category>{CATEGORY}</category>
    </categories>
    <person orcid="{ORCID}" role="curator">{CURATOR_NAME}</person>
    <person role="designer" url="{DESIGNER_URL}">{DESIGNER_NAME}</person>
    <name>{ICON_NAME}</name>
    <description>{DESCRIPTION}</description>
    <references>
        <reference>
            <db>{REFERENCE_NAME}</db>
            <id>{REFERENCE_ID}</id>
        </reference>
    </references>
    <synonyms>
        <synonym>{SYNONYM}</synonym>
    </synonyms>
</metadata>
```

Please follow these steps:

1. Create a new file using your favorite text editor (TextEdit, NotePad, etc.).
2. Copy and Paste the text above into your file.
3. Edit the file and:
 1. Fill {CATEGORY} with one or more of the suggested categories: cell_element, cell_type, compound, human_tissue, protein, receptor or transporter. We accept one or more different categories as we think they are just a guide that helps our users to find the element they need.
 2. Fill {ORCID} and/or {URL} for the curator and designer roles. This is how to find ORCID. In case both URL and ORCID are provided, the link for that person will point to the URL.
 3. Fill {CURATOR_NAME} and {DESIGNER_NAME} as the curator and designer name, respectively. If the name is the same, please fill it with both same information.
 4. Fill {DESCRIPTION} with any information relevant to describe the component. Please be brief.
 5. Fill {REFERENCE_NAME} for the name of the resource you have used to find this component (e.g.:CHEBI, UNIPROT, GO ...). In case of not finding a suitable reference, please delete this line.
 6. Fill {REFERENCE_ID} as it appears in the resource's URL (e.g.: CHEBI:1234, Q1234, GO:1234 ...). In case of not finding a suitable reference, please delete this line.
 7. Fill {SYNONYM} as any synonyms or alternative names your component might be known of. In case of not finding a suitable synonym, please delete this line.
4. Save the file using the same as the element but make sure to use .xml as its extension. For example if your graphic element is stored with the name "Element.svg" then your metadata file should be named as "Element.xml".
5. Make sure to submit your metadata file along with the graphics file.

Export Settings

Currently, every element in the Reactome Library is available in 3 different formats, SVG, EMF and PNG. Therefore, you are kindly requested to export your work into these formats and include them in your submission.

For PNG files we suggest to export with 300 DPI and transparent background.

How to submit your work

To include your graphic elements in the Reactome Library, you need to contact our helpdesk (help@reactome.org). We will then review your submitted files and, depending on the outcome, we will include them in our Library.

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